GAME THEORY

ADVERSARIAL SEARCH

"In which we examine the problems that arise when we try to plan ahead in a world where other agents are planning against us".

- In this part we cover competitive environments, in which the agents' goals are in conflict, giving rise to adversarial search problems—often known as games.

- Goal:
  Model and analyze (using mathematical tools) situations of interactive decision making.

- Ingredients:
  - Several decision makers (called players)
  - All with different goals
  - The decision of each players impacts the outcome for all

- Wide range of applicability

  "it is a context-free mathematical toolbox"

- Social science, theoretical economics, Political science, Biology and others.

- Game Theory is a branch of mathematics used to model the strategic interaction between different players in a context with predefined rules and outcomes.

- These games can be classified
  - Number of players: 1 or 2 or 3 or . . .
  - Type of interactions: simultaneous or sequential
  - Maximal length of a play: finite or infinite
  - Type of information: perfect or imperfect
  - Presence of randomness: deterministic or probabilistic
  - Type of payoff: boolean or quantitative

The followings are imported from: https://towardsdatascience.com/game-theory-in-artificial-intelligence-57a7937e1b88

Game Theory can be divided into 5 main types of games:

- **Cooperative vs Non-Cooperative Games:** In cooperative games, participants can establish alliances in order to maximise their chances to win the game (eg. negotiations). In non-cooperative games, participants can't instead form alliances (eg. wars).
- **Symmetric vs Asymmetric Games:** In a symmetric game all the participants have the same goals and just their strategies implemented in order to achieve them will determine who wins the game (eg. chess). In asymmetric games instead, the participants have different or conflicting goals.
- **Perfect vs Imperfect Information Games:** In Perfect Information games all the players can see the other players moves (eg. chess). Instead, in Imperfect Information games, the other players' moves are hidden (eg. card games).
- **Simultaneous vs Sequential Games:** In Simultaneous games, the different players can take actions concurrently. Instead in Sequential games, each player is aware of the other players' previous actions (eg. board games).
- **Zero-Sum vs Non-Zero Sum Games:** In Zero Sum games, if a player gains something that causes a loss to the other players. In Non-Zero Sum games, instead, multiple players can take benefit of the gains of another player.

Single-move games

- Where all players take action simultaneously and the result of the game is based on this single set of actions. (Actually, it is not crucial that the actions take place at exactly the same time; what matters is that no player has knowledge of the other players' choices.)

- A single-move game is defined by three components:

- **Players** or agents who will PLAYER be making decisions. Two-player games have received the most attention, although n-player games for n > 2 are also common.
- **Actions** that the players can choose. The players may or may not have the same set of actions available.
- **A payoff function** that gives the utility to each player for each combination of actions by all the players. For single-move games the payoff function can be represented by a matrix.

Payoff Matrix

Two-finger Morra

There are two players, O and E, simultaneously display one or two fingers. Let the total number of fingers be f. If f is odd, O collects f dollars from E; and if f is even, E collects f dollars from O.

	O: one	*O: two*
E: one	$E = +2, O = -2$	$E = -3, O = +3$
E: two	$E = -3, O = +3$	$E = +4, O = -4$

(2 , -2)	**(-3 , 3)**
(-3 , 3)	**(4 , -4)**

Prisoner's dilemma

	$Alice{:}testify$	$Alice{:}refuse$
$Bob{:}testify$	$A = -5, B = -5$	$A = -10, B = 0$
$Bob{:}refuse$	$A = 0, B = -10$	$A = -1, B = -1$

(-5 , -5)	**(-10 , 0)**
(0 , -10)	**(-1 , -1)***

Nash Equilibrium

7,6	5,5
4,5	6,4

7,6*	5,5
4,5	6,4

8,7	4,6
6,5	7,8

8,7*	4,6
6,5	7,8*

9,9	1,10
10,1	2,2

9,9	1,10
10,1	2,2*

6,8	4,7
7,6	3,7

6,8	4,7
7,6	3,7

Search versus Games

Search –no adversary

- **Solution is a path from start to goal, or a series of actions from start to goal.**
- **Heuristics search techniques (informed search) can find optimal solution.**

Games –adversary

- **Solution is strategy (Strategy specifies move for every possible opponent reply).**

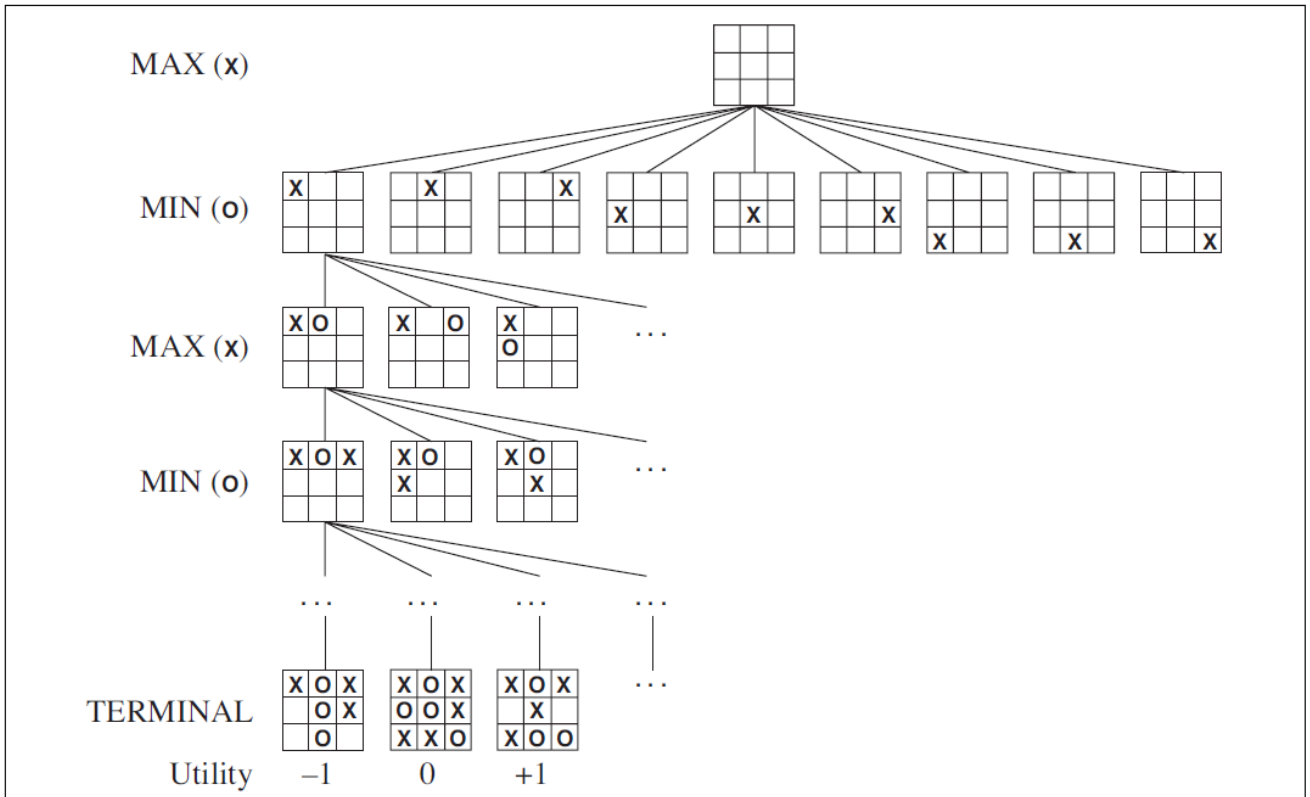
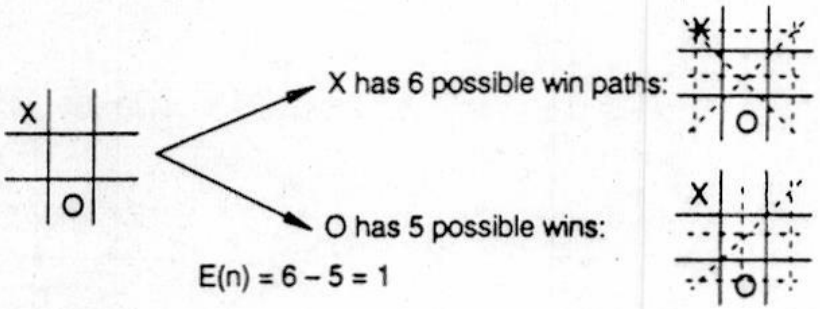
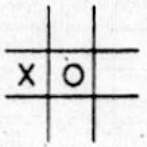


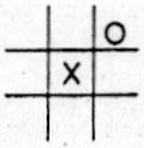
Figure 5.1 A (partial) game tree for the game of tic-tac-toe. The top node is the initial state, and MAX moves first, placing an X in an empty square. We show part of the tree, giving alternating moves by MIN (O) and MAX (X), until we eventually reach terminal states, which can be assigned utilities according to the rules of the game.





X has 4 possible win paths;
O has 6 possible wins

E(n) = 4 – 6 = –2



X has 5 possible win paths;
O has 4 possible wins

E(n) = 5 – 4 = 1

Heuristic is E(n) = M(n) – O(n)

where M(n) is the total of My possible winning lines

O(n) is total of Opponent's possible winning lines

E(n) is the total Evaluation for state n

Figure 4.16 Heuristic measuring conflict applied to states of tic-tac-toe.

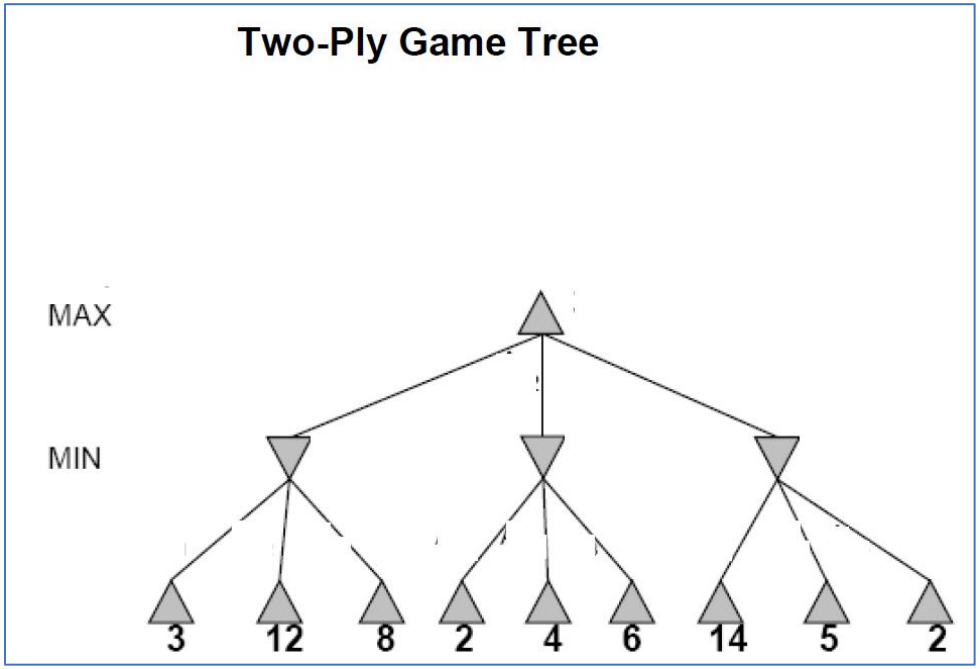
Optimal Decisions in Games

In a normal search problem, the optimal solution would be a sequence of actions leading to a goal state—a terminal state that is a win. In adversarial search, MIN has something to say about it. MAX therefore must find a contingent strategy, which specifies MAX's move in the initial state, then MAX's moves in the states resulting from every possible response by MIN, then MAX's moves in the states resulting from every possible response by MIN to those moves, and so on.

Two- Ply Game

- **We consider games with two players, whom we call MAX and MIN.**
- **The tree in this game is one move deep, consisting of two half-moves, each of which is called a ply**
- **How do we search this tree to find the optimal move?**

The Minimax Algorithm



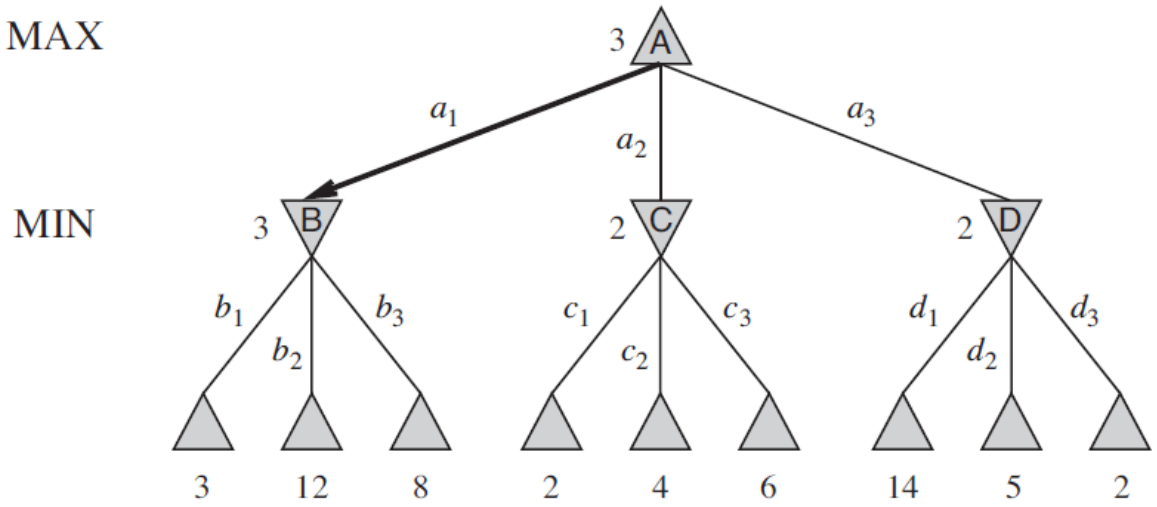
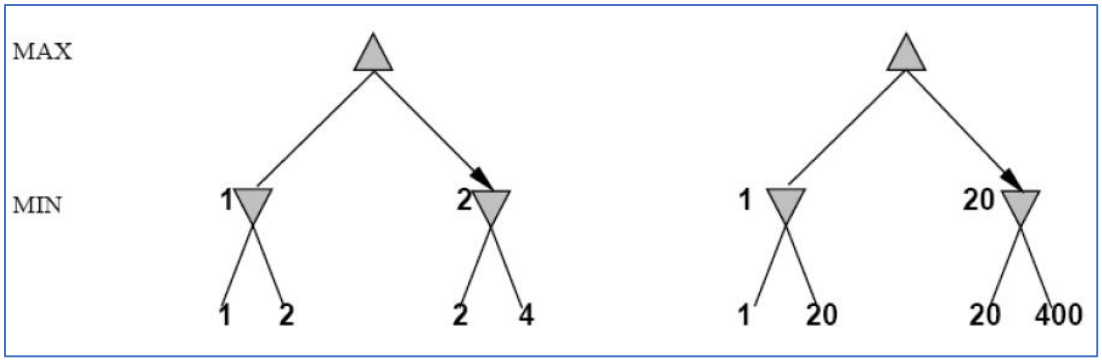


Figure 5.2 A two-ply game tree. The △ nodes are "MAX nodes," in which it is MAX's turn to move, and the ▽ nodes are "MIN nodes." The terminal nodes show the utility values for MAX; the other nodes are labeled with their minimax values. MAX's best move at the root is a_1, because it leads to the state with the highest minimax value, and MIN's best reply is b_1, because it leads to the state with the lowest minimax value.


MAX
MIN
1
2
1
2
2
4
1
20
1
20
20
400

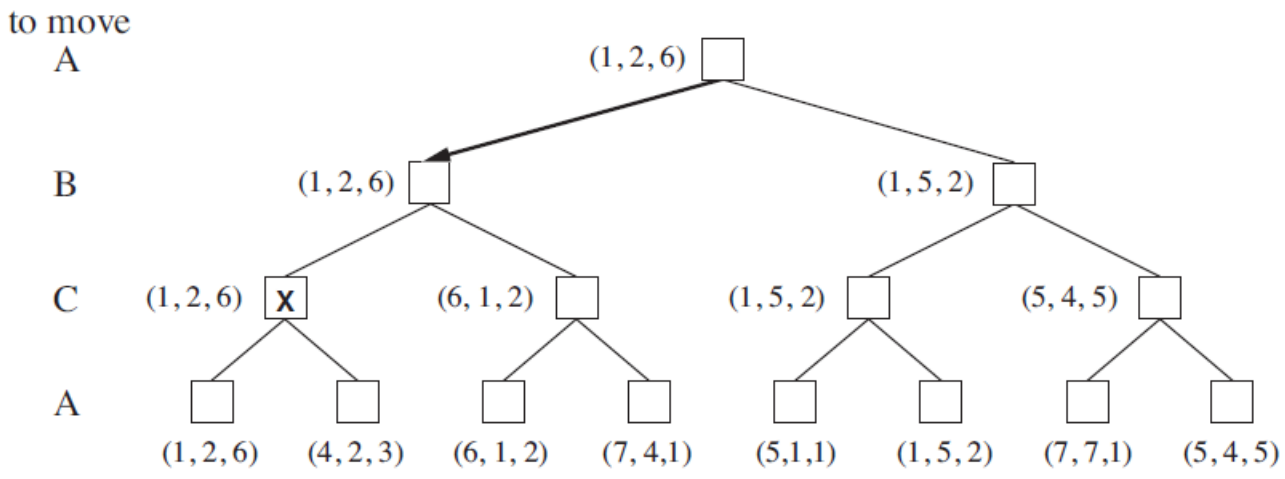


Figure 5.4 The first three plies of a game tree with three players (A, B, C). Each node is labeled with values from the viewpoint of each player. The best move is marked at the root.

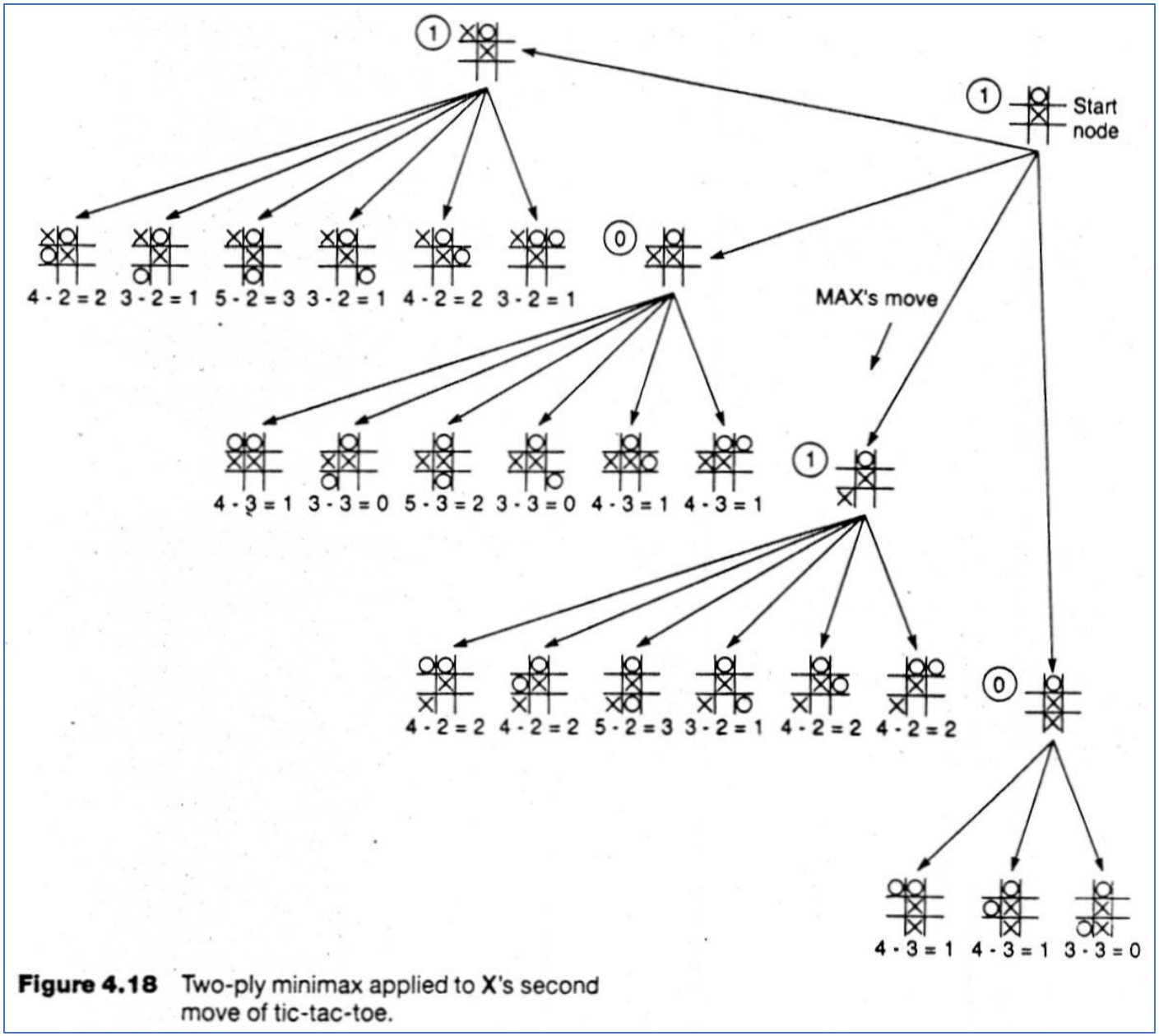
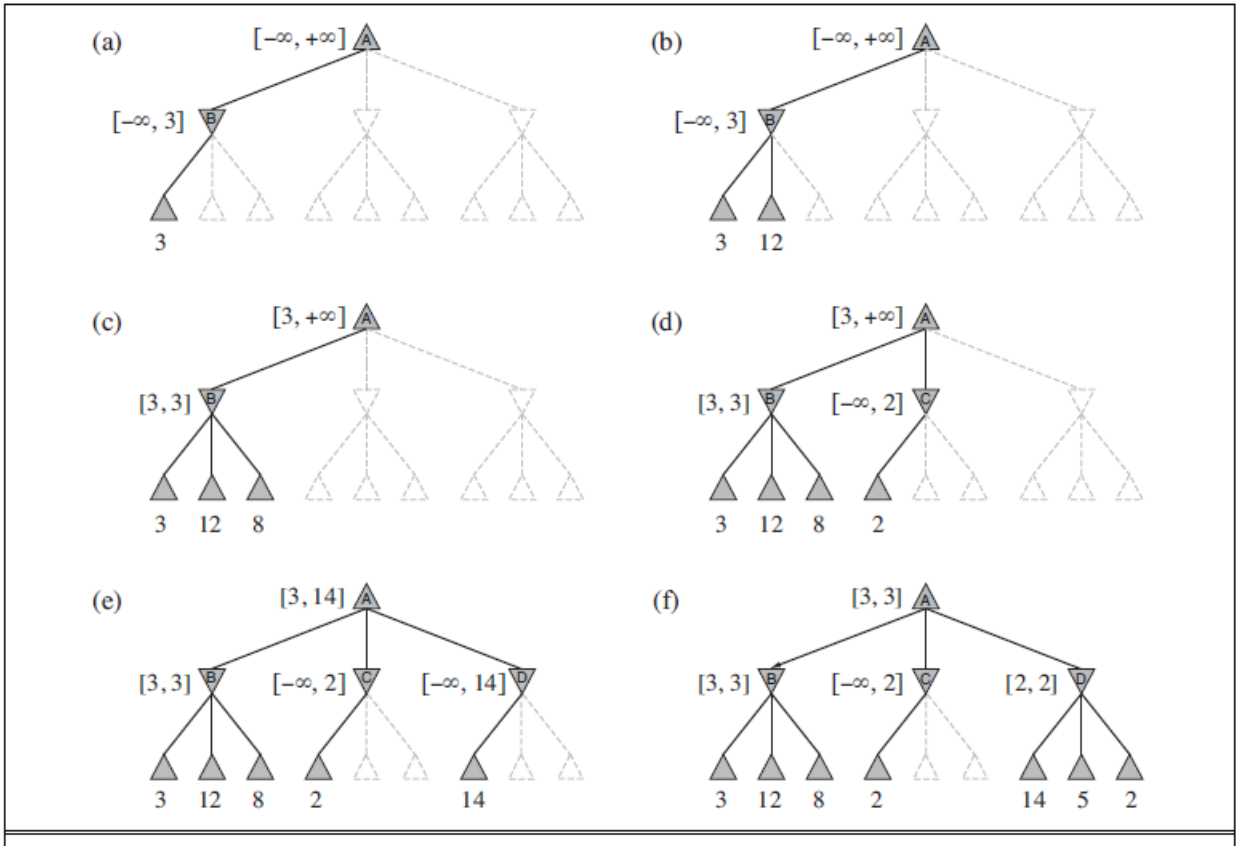


Figure 4.18 Two-ply minimax applied to X's second move of tic-tac-toe.

Alpha-Beta Pruning algorithm



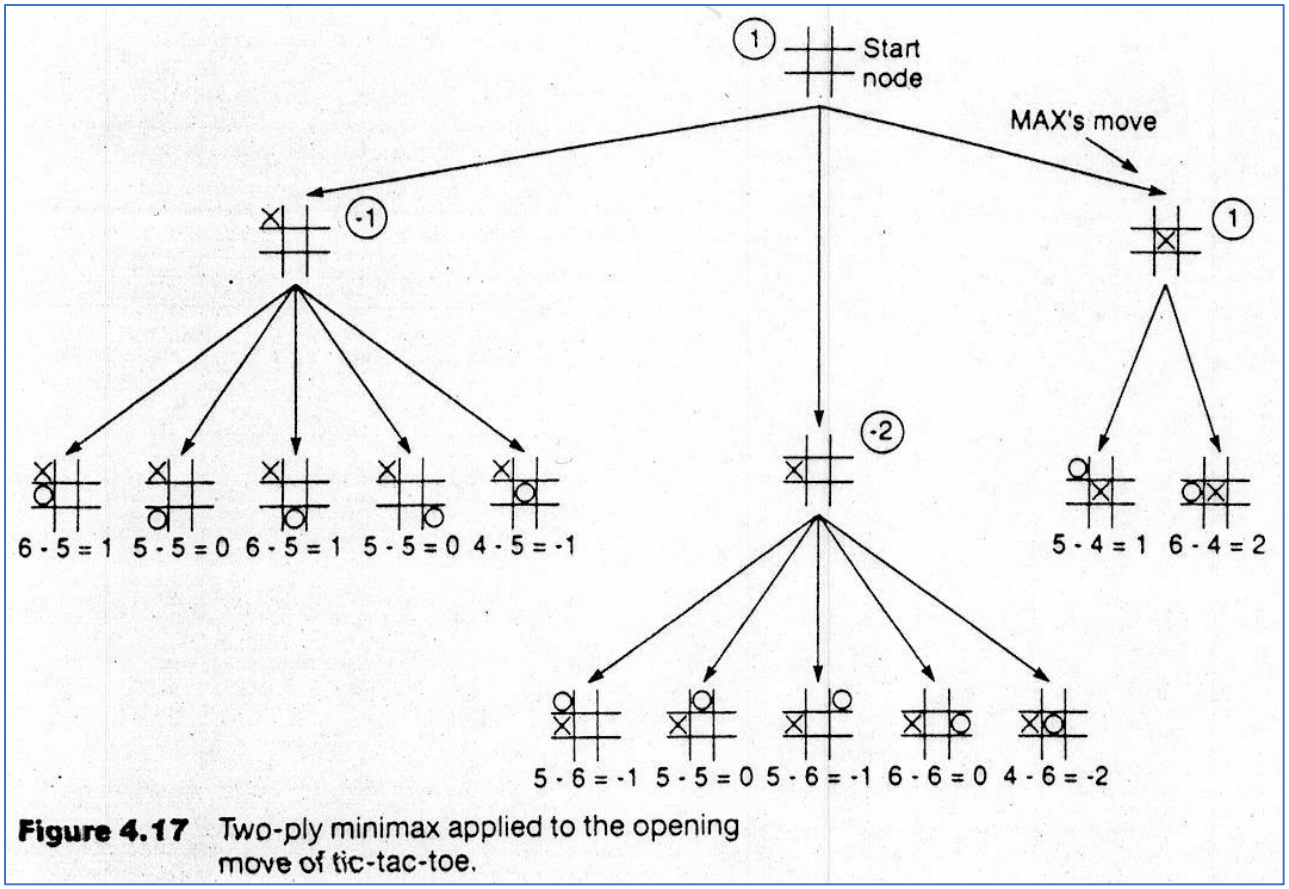


Figure 4.17 Two-ply minimax applied to the opening move of tic-tac-toe.